COSC2820 Advanced Programming for Data Science

COSC 2820/2815
Assignment 2: NLP Web-based Data Application
Milestone II: Web-based Data Application

Assessment Type	Individual assignment. Submit online via Canvas→Assignments→Assignment 2→ Milestone II: Web-based Data Application. Marks awarded for meeting requirements as closely as possible. Clarifications/updates may be made via announcements/relevant discussion forums.
Due Date	Week 13, Monday 3rd of June 9:00 am
Marks	10

1. Overview

In Milestone I you built various machine learning models using different document vector representations to classify job adverts. In Milestone II, you will develop a job search website based on the Flask web framework. The website allows job seekers to browse existing job adverts and lets employers create new job adverts. This website will make use of one machine learning model that you trained in Milestone I. It will be used to recommend job categories for employers to assign to newly created job adverts. This feature will help to reduce human error, increase the job exposure to relevant candidates, and improve the user experience of the job seeking website.

2. Learning Outcomes

This assessment relates to following learning outcomes of the course:

- CLO 5: Document and maintain an editable transcript of the data pre-processing pipeline for professional reporting;
- CLO 6: Build small to medium scale data-driven applications using a Web development framework.

3. Assessment details

In this milestone, you are required to build a job seeking website using Flask.

Job Data

To test and demonstrate the functionality of the website, you will need some job data. In this milestone, you can use data, including the job information and descriptions, given in Milestone I. If you want, you can also create any additional artificial data (e.g. images) for the purpose of display.

Website Layout and Design

The layout and design of the website is flexible. You can refer to the exercises in the material from Week 10 and 11, or the Job Browsing Page and the Employer site of the seek.com.au

Minimum Functional Requirements

There are minimum functional requirements of the developed website for job hunters and employers, respectively. Feel free to include other functionalities that suit.

Functionality for Job seekers

Job Search

The Job hunting website will allow job hunters to effectively search for job listings that are of their interest. The search could be based on keywords. Upon user entering a keyword string, the develop system should return a message saying how many matched job advertisements, and it also returns a list of job advertisement previews that are relevant to the keyword string. When users click on a job ad preview, they can see the full description of the job. You need to design a search algorithm that:

- support to search keyword strings in similar forms. For example, if users enter the keyword strings "work" or "works" or "worked", the search results from these two keyword strings will be the same.
- Apart from the above requirement, you have the flexibility to use simple string matching based methods, or any other language models, or build your own models to evaluate the relevance of a job advertisement to the entered keyword string.

Functionality for Employers

Create New Job Listing

The Job seeking website allows employers to create new job listings. When creating a new job listing employers can enter information such as title, description, salary, etc.
When a new job advert is created, using the job description (and/or job title), the website should recommend a list of categories for selection to assign to the job advert. The employer should be able to select other categories if the recommendation does not suit, that is, they can override the categories suggested by the website (you may refer to the employer site at seek.com.au as an example). Upon confirmation, the created job advert should be included in the job data and be accessible via URL and relevant search.
You have the flexibility to choose the language model and the classification model for the job category recommendation task.
You do not need to create any employer login for the job website.

4. Marking Guidelines

Marking Criteria

Marking of this milestone will consider both the design and implementation of the website.

Mark Allocations

- Design & Layout [20%]
- Implementation of functional requirements [total 80%]
    - Job Searching (40%)
    - Create a New Job Listing (40%)

5. Submission

The final submission of this milestone must include:

- The Python source code
- A README.txt file with your student number and any important details.
- Any other files/data that are required to locally run the website. Therefore, make sure all necessary files to run your code are included in your submission. If these files are greater than 50MB package them in a zip file and upload them to OneDrive. Provide OneDrive URL in the README.txt file.
- Package these file files in a folder, named with your student id, zip the folder with the same name (i.e., **<student_number>.zip**) and upload Canvas.
- All source code will be passed to a plagiarism checker to ensure academic integrity.

Assessment declaration:
When you submit work electronically, you agree to the assessment declaration:
https://www.rmit.edu.au/students/student-essentials/assessment-and-exams/assessment/assessment-declaration

Late Submission Penalty
Late submissions will incur a 10% penalty on the total marks of the corresponding assessment task per day or part of day late. Submissions that are late by 5 days or more are not accepted and will be awarded zero, unless special consideration has been granted. Granted Special Considerations with a new due date set more than 2 weeks after the original due will automatically result in an equivalent assessment in the form of a practical test with interview, assessing the same knowledge and skills of the assignment (location and time to be arranged by the instructor). Please ensure your submission is correct (all files are there, compiles etc), re-submissions after the due date and time will be considered as late submissions.

6. Academic integrity and plagiarism (standard warning)

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarized, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods,
- provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites.

If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviors, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on our policies and procedures, please refer to https://www.rmit.edu.au/students/student-essentials/rights-and-responsibilities/academic-integrity